

Quality Control Manual — SteelForge Manufacturing

1. Purpose and Scope

This Quality Control Manual establishes the procedures, standards, and responsibilities for ensuring product quality across all SteelForge Manufacturing production lines. It covers raw material inspection, in-process quality checks, final product testing, non-conformance reporting (NCR), corrective and preventive actions (CAPA), and compliance with ISO 9001:2015 and IATF 16949 automotive quality management standards.

All production personnel, quality inspectors, and line supervisors are required to follow the procedures outlined in this manual. Deviations must be documented via the Non-Conformance Report (NCR-Form-101) and escalated within 24 hours.

2. Incoming Material Inspection

All raw materials received at the loading dock must undergo inspection within 4 hours of delivery. The Quality Lab team uses Acceptance Quality Level (AQL) sampling per ANSI/ASQ Z1.4 standard. Critical materials such as high-tensile steel coils, aluminum billets, and copper wire stock require 100% dimensional verification using coordinate measuring machines (CMM). Chemical composition is verified via optical emission spectroscopy (OES) against material test certificates (MTC).

Supplier performance is tracked through the Supplier Scorecard system. Suppliers with a quality rating below 85% are placed on probation and require corrective action plans within 30 days. Repeat offenders (three consecutive months below threshold) are escalated to the Approved Vendor List (AVL) committee for potential removal.

Material	Spec	Test Method	Frequency	Accept Criteria
Steel Coil HR	ASTM A36	Tensile + CMM	Every lot	Yield $\geq$ 250 MPa
Aluminum 6061-T6	AMS 4027	Hardness + OES	Every lot	Brinell $\geq$ 95
Copper Wire C110	ASTM B3	Conductivity	1 per 5 lots	$\geq$ 100% IACS
Stainless 304	ASTM A240	Corrosion test	Every lot	Pass 48h salt spray
Titanium Gr5	AMS 4928	Ultrasonic + CMM	100%	No defects > 0.5mm

3. In-Process Quality Control

Statistical Process Control (SPC) is mandatory on all CNC machining, stamping, and welding operations. Control charts (X-bar and R charts) are maintained for critical dimensions. Process capability indices (Cpk) must be ≥ 1.33 for standard features and ≥ 1.67 for safety-critical dimensions. Real-time SPC data is collected via shop-floor terminals connected to the MES (Manufacturing Execution System) and monitored by Quality Engineers on the SPC Dashboard.

First Article Inspection (FAI) is required for all new part numbers, engineering change orders (ECO), and after any tooling change. FAI reports follow AS9102 format and must be approved by the Quality Manager before production release. All measurement equipment is calibrated per ISO/IEC 17025 with calibration intervals not exceeding 12 months.

4. Final Product Testing and Release

Finished goods undergo a battery of tests before shipment release. Mechanical testing includes tensile, impact (Charpy V-notch), and fatigue testing per customer specifications. Dimensional inspection uses CMM with GD&T; callouts per ASME Y14.5-2018. Surface finish is verified using profilometers (Ra values). All test results are recorded in the Enterprise Quality Management System (EQMS) and linked to the lot/batch traceability record.

Products failing final inspection are quarantined in the Material Review Board (MRB) area. The MRB team (Quality Manager, Production Manager, Engineering Lead) reviews disposition options: rework, use-as-is (with customer concession), or scrap. All MRB decisions are documented and require customer approval for safety-critical components.

5. Non-Conformance and CAPA Process

Non-conformances are classified as Minor, Major, or Critical. Critical NCRs trigger an immediate production stop and 8D problem-solving process. Root Cause Analysis (RCA) uses 5-Why, Fishbone (Ishikawa), and Fault Tree Analysis (FTA) methods. Corrective actions must be implemented within 15 business days for Critical NCRs, 30 days for Major, and 60 days for Minor. Effectiveness verification is conducted 90 days after implementation.

NCR Severity	Response Time	RCA Deadline	CAPA Deadline	Verification
Critical	Immediate stop	48 hours	15 business days	90-day audit
Major	24 hours	5 business days	30 business days	90-day review
Minor	72 hours	10 business days	60 business days	Next scheduled audit